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A Scandinavian Challenge, a US Response: Methodological Assumptions in Scandinavian and US Prototyping Approaches

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ABSTRACT

In the early 1980s, Scandinavian software designers who sought to make systems design more participatory and democratic turned to prototyping. The “Scandinavian challenge” of making computers more democratic inspired others who became interested in user-centered design; information designers on both sides of the Atlantic began to employ prototyping as a way to encourage user participation and feedback in various design approaches. But, as European and North American researchers have pointed out, prototyping is seen as meeting very different needs in Scandinavia and in the US. Thus design approaches that originate on either side of the Atlantic have implemented prototyping quite differently, have deployed it to meet quite different goals, and have tended to understand prototyping results in different ways.

These differences are typically glossed over in technical communication research. Technical communicators have lately become quite excited about prototyping’s potential to help design documentation, but the technical communication literature shows little critical awareness of the methodological differences between Scandinavian and US prototyping. In this presentation, I map out some of these differences by comparing prototyping in a variety of design approaches originating in Scandinavia and the US, such as mock-ups, cooperative prototyping, CARD, PICTIVE, and contextual design. Finally, I discuss implications for future technical communication research involving prototyping.

Categories and Subject Descriptors

K.4.3 [Computers and Society]: Organizational Impacts – automation, computer-supported collaborative work, employment.

General Terms

Management, Documentation, Design, Human Factors, Theory.

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Keywords

CARD, collective resource approach, contextual design, cooperative prototyping, democracy, participatory design, PICTIVE, prototyping, Scandinavian design, user-centered design

1. INTRODUCTION

Sweden, 1981: Pelle Ehn and collaborators at the Royal Institute of Technology in Stockholm and the University of Aarhus, Denmark began the UTOPIA project. This project was a collaboration between academics and a typesetter’s union that was meant to give the typesetters a say in how computer technology would be implemented at a newspaper. As committed Marxists, the academics saw the workplace as a continuing conflict between capital (employers) and labor (employees); in this struggle, the employers had traditionally had the upper hand. Backed by the union, the academics sought to provide a proactive way for workers to implement industrial democracy in their workplace. One important part of that strategy was *mock-ups*. Mock-ups provided a common language for academics and workers; a way to iteratively try out solutions; a way for workers to democratically develop consensus on what they wanted from a computer system; and a path for further training and empowering workers in systems design.

The United States, 1998: Hugh Beyer and Karen Holtzblatt of InContext Enterprises published a book describing their design approach, contextual design. The book supplements the training sessions that InContext provides (typically over \$1000 for a three-day training session) and reflects the consulting work that InContext does for companies. Aimed at managers and IT personnel in US companies, the book describes contextual design as a “customer-centered” design approach that revolves around observing aspects of employees’ work, developing work models based on the observations, and basing new system designs on the work models. Once these systems have been outlined and prototyped, designers hold one-on-one “prototyping sessions” with workers to get feedback and gather ideas. The workers, although they are termed “co-designers,” are emphatically not designing the system using the prototypes: “It’s their job to do their job, not design systems” [5].

The two prototyping approaches are startling in their differences. Whereas UTOPIAN mock-ups are based on Marxist theory, contextual design is emphatically capitalist. In the UTOPIA project, the academics were called in by the union and were financed with grant money from the government; InContext is

hired by managers and paid from company coffers. UTOPIAN mock-ups are meant to draw workers into the design process and make them designers, *not* just to test, ratify, or tweak the work of external designers; contextual design's prototyping component is designed to disrupt work as little as possible and to relieve workers of the additional work of design, so it encourages them to make minor changes and suggestions – but explicitly *not* to become designers themselves or to disrupt the division of labor established by the design team.

What is so striking about this contrast is that when Beyer and Holtzblatt describe their prototyping component, they cite the Scandinavian researchers who were involved in the UTOPIA project. How did this shift come about? How did the “Scandinavian challenge” of workplace democracy [8; compare with 18] become translated so radically as it migrated to the shores of the United States?

To make sense of the transformation, I look to the distinction between democratic and functional empowerment. Jeanette Blomberg, Lucy Suchman, and Randy Trigg [10] draw on Andrew Clement's work [19] to distinguish between the two sorts of empowerment: “Put simply, functional empowerment holds work groups accountable for the results of tasks, and in return gives them a degree of power over *how* to execute the tasks. Democratic empowerment ideally gives workers a decision-making role in operational planning as well as organizational and technological change” (281). In these terms, UTOPIAN mock-ups reflect a desire for *democratic* empowerment, in which workers make decisions along with management; contextual design strives for *functional* empowerment, in which workers are better able to perform their jobs.

In this paper, I review a few of the many overlapping prototyping approaches that have sprung up in the wake of UTOPIA – mock-ups, cooperative prototyping, PICTIVE, CARD, and contextual design – and examine how each approach compares to the Scandinavian challenge. (See [41] for a more comprehensive survey of approaches.) I sum up by describing how technical communicators might use this understanding of prototyping when selecting and using prototyping approaches.

2. BACKGROUND: THE SCANDINAVIAN CHALLENGE

The “Scandinavian challenge” was based in the Collective Resources Approach (CRA), sometimes known as the critical tradition of Scandinavian information technology [2, 6] or simply the Scandinavian approach. In 1971, Scandinavian academics and unions, led by Kristen Nygaard, began a series of projects aimed at empowering labor in its struggle with management, particularly in terms of the introduction of new technologies [43, 2]. This struggle was framed in terms of democracy, based on Marx's observation that in a capitalist society, democracy stops at the factory gates – democracy in the workplace was counter to management's goals of higher profits for less pay. Nygaard and his collaborators chose to work with unions in the manufacturing industry. At the time, Scandinavia was about 90% unionized, so unions had considerable power, yet unions were not well equipped to proactively guide the introduction of technologies.

Nygaard's work was overtly political; he argued that research is always political, but his work was overt because it advocated the case of labor rather than management [43, 2]. (Indeed, later

scholars in this tradition saw the political researcher as an “emancipator” [7]). Others built on this work, and eventually the Collective Resource Approach took shape.

CRA projects dealt with how new technologies were introduced into workplaces and how they devalued workers' traditional craft skills, deskilling jobs, removing workers' ability to make decisions, and intensifying work. At first CRA projects involved educating workers about technology and organizing resistance to technologies that reduced industrial democracy. But in the UTOPIA project, led by Pelle Ehn, the academics began to realize that workers had to be empowered to provide practical alternatives to management-based initiatives. That is, workers had to be able to describe a computer system that could automate work while still valuing their craft skills and upholding their autonomy. The problem was that the workers had no experience in systems design. Thus they could not begin to speculate on how to build such a system. So the UTOPIA team began experimenting with a range of techniques for discussing and exploring the possibilities, including mockups and other low-fidelity prototypes, but also “future workshops” and organizational toolkits [13].

The UTOPIA project was a critical success, if not a commercial one. Although the project failed to produce a working system, it did produce a design approach and a range of techniques for participatory design work. Based on UTOPIA and other projects that came after it, the Scandinavians issued the “Scandinavian challenge” in a 1987 book: develop and use design approaches that encourage industrial democracy [8].

Many took up this challenge, particularly in Great Britain and the United States [37]. But, particularly in the US, the Scandinavian approach lost some things in the translation. For instance, in Scandinavia unions had been considered a key ingredient for participatory design. Pelle Ehn called unions “the only real social ‘carrier’ of the ideals of work-oriented design” [21]. That made sense in a region with a workforce that was 90% unionized, but in the US, far fewer workers were unionized – after a postwar high of 40%, unionization has dropped to around 15% [32; see also 3, 7, 49]. Although some in the US were still hopeful that unions could be a driving force in US participatory design [25], others discounted that possibility and sought other avenues for empowering workers [26].

Nevertheless, US academics and researchers sought to find other ways to apply the Scandinavian approach, and in doing so, they modified its techniques and goals and added their own.

3. “UTOPIAN” MOCK-UPS (SCANDINAVIA)

3.1 Description

The use of mock-ups – low-fidelity prototypes that serve to introduce new technologies – was one of the techniques pioneered in the UTOPIA project, and it quickly became a central technique in participatory design. It was based on the *tool perspective*, the idea that “computer support is designed as a collection of tools for the skilled worker to use. The tool perspective takes the *work process* as its origin rather than data or information flow. This means: *not* detailed analysis, description, and formalization of qualifications, *but* the development of professional education based on the skills of professionals; *not* information flow analysis and systems description *but* specification of tools” [13, p.261].

The tool perspective involved recognizing the skilled craft that workers had developed over the years so that new tools would *fit into* that craft, rather than disrupting it: “a tool is developed as an extension of the accumulated knowledge of tools and materials within a domain” [13, p.261; see also 21, p.339-40]. In contrast to design approaches favored by management that served Tayloristic goals (deskilling, work intensification), the tool perspective involved building “computer-based tools by which the craftsman [sic] can still apply and develop original skills” [13, p.261; see 21, p.34; 23, p.34-38].

In the UTOPIA project, mock-ups became important in bridging the experience of researchers and workers. The researchers had begun by using traditional design description methods such as scenarios and data flow. But the graphical workers had difficulty understanding these unfamiliar notations, so the researchers and workers turned to mock-ups [13, p.263]. “The principle behind the mock-up was simple. Using sheets of paper, matchboxes, some plywood, etc., one ‘builds’ a workstation with a ‘high-resolution display,’ a ‘mouse,’ etc.” [21, p.335]. Then, under the researcher’s guidance, workers used the mock-ups to simulate work [22].

Mock-ups had several advantages. One was that they provided an easy way for researchers to introduce new technologies (such as laser printers, which existed only in advanced research labs during the time of the UTOPIA project; see [22, p.172]. Another was that workers were able to explore how these new technologies might fit into their existing tacit knowledge about their craft, allowing them and the researchers to refine the design in ways that made use of their existing skills. They were also cheap – a requirement in this project, which ran on quite limited resources, and also a boon to other projects supported by unions rather than management. Most importantly, mockups provided a “language game” that made sense to all participants – a way for researchers and workers to pool their expertise and share design responsibility for the project [22, p.176-77]. This “language game” meant that workers could even modify the prototypes to better suit their tacit knowledge of their craft [22, p.183]. The new design, then, would be co-designed with the workers and would extend their craft skills rather than doing away with them.

3.2 Comparison to the Scandinavian Challenge

As they were used by Ehn and his colleagues in the UTOPIA project, mock-ups were an important component of the Scandinavian challenge. They were explicitly developed to promote democratic empowerment: they provided a common language for designers and workers and they allowed workers to co-design the system, giving them an avenue to control the introduction of new technologies in ways that were beneficial to them. Additionally, the workers who participated in mock-up sessions were *representative users* in the democratic sense, the sense that they were chosen by the local union to represent the other members of the union.

Democratic empowerment was the ultimate goal, but it was achieved partially through functional empowerment: mock-ups encouraged an exploration of workers’ tacit craft knowledge and modeled the new tools on that craft knowledge (some might say that it *leveraged* that knowledge). Certainly these changes resulted in a system that workers found functionally empowering and

easier to use, but the mock-ups’ primary goal was democratic empowerment. The UTOPIA researchers explicitly rejected *efficiency* as a goal, recognizing that it often equals work intensification and deskilling.

On the other hand, Gro Bjerknes and Tone Bratteig have pointed out that UTOPIA failed to achieve its democratic aims, largely because the research and design were done on behalf of a single union rather than the host of unions at the site: “research on behalf of one union does not necessarily contribute to a more democratic working life” [7, p.78]. They concluded, “We consider the UTOPIA project ... as a support to graphical workers at the expense of women and unskilled men in the composers’ room ... the UTOPIA project has not contributed to the sort of democracy in which all stakeholders have a voice in the design of a new computer system” (78).

4. COOPERATIVE PROTOTYPING (SCANDINAVIA)

4.1 Description

Cooperative prototyping developed as a part of the cooperative design approach that emerged in the wake of UTOPIA ([17]; see [23] for an early volume on cooperative design and [33] for a later perspective). Introduced in [16], cooperative prototyping was intended to be “a cooperative activity between users and designers, rather than an activity of designers utilizing users’ more or less articulated requirements” (p.200).

Cooperative prototyping was based on the “tool perspective” first described in the UTOPIA project. Like UTOPIA mock-ups, prototypes in the cooperative prototyping technique served as a common design language for designers; provided a way to try out common solutions before committing to them; and allowed the two parties to cooperatively design tools that fit into the craftwork of the workers. Importantly, the latter meant that researchers could explore workers’ tacit knowledge, something that became a major theme in cooperative prototyping literature.

This exploration of tacit knowledge happened primarily through examining *breakdowns*. In a cooperative design session, users would “experience a fluent work-like situation with a future computer application; that is, users’ current skills must be brought into contact with new technological possibilities” [16, p.200]. As the workers used the prototype, they sometimes encountered breakdowns, points at which they shift their focus from the work to the tool because the tool is behaving unexpectedly (cf. [12]). At these points, “users and designers can analyze the situation and discuss whether the breakdown occurred because of the need for training, a bad or incomplete design solution, or for some other reason” [16, p.200]. If a breakdown was caused by the design, users and designers could modify the prototype on the spot, exploring other design solutions that might minimize breakdowns. The result was ideally a modified prototype that was the result of a user-designer collaboration and that complemented the craft skills of the workers rather than deskilling them. Indeed, cooperative prototyping saw prototypes as part of users’ learning, not just feedback to designers [17, p. 170].

Unlike mock-ups, cooperative prototyping focused almost exclusively on computer applications and involved rapid development tools (such as HyperCard; see [15]). Since these prototypes were automated on a computer screen, they were

interactive, providing users with a better understanding of how they might work with a finished product. In addition, simple changes to online prototypes could be easier than changes to mock-ups, meaning that workers and designers could collaboratively redesign on the fly [16]. Cooperative prototyping became more flexible in collaboratively designing computer applications, but at the expense of shutting out other sorts of design work.

4.2 Comparison to the Scandinavian Challenge

Cooperative prototyping was still concerned with democratic empowerment, but it assumed that democratic empowerment would necessarily follow from functional empowerment [7]. Consequently, cooperative prototyping's focus narrowed to individual workers and tools, and unions were no longer seen as a vital part of the equation [27, p.14]. Most notably, the Marxist theme of capital vs. labor became muted. In one striking example, Bodker and Gronback counseled against bringing middle managers in as participants, not because they would co-opt the process and disempower workers, but because they simply didn't have the tacit knowledge that their workers do [16, p.210]. Finally, researchers began to make arguments for cooperative prototyping that appealed to management's interests and that had little to do with democratic empowerment: avoiding unnecessary work and lowering costs [16, pp.215-16]. Although cooperative prototyping emerged from the same Scandinavian milieu and indeed the same researchers who had issued the Scandinavian challenge, it had begun to lose its focus on the ideals embedded in the challenge.

5. PICTIVE AND CARD (US)

5.1 Description

As the "Scandinavian challenge" began to be publicized in the US, many academics and researchers became interested in taking up the challenge. Two of the more notable results are PICTIVE (Plastic Interface for Collaborative Technology Initiatives through Video Exploration) and CARD (Collaborative Analysis of Requirements and Design)¹, two complementary approaches developed by Michael Muller and colleagues.

In developing PICTIVE, Muller related the Scandinavian commitment to workplace democracy to US democratic ideals through his admittedly "glib" slogan of "no mechanization without representation," a paraphrase of the phrase from the US Declaration of Independence, "no taxation without representation" [36, p.391]. Muller wanted "to enfranchise users in the design of systems that would have impact on their work-lives" (p.391).

In contrast to the Scandinavian approach, however, Muller looked beyond unions for his representatives. He did sometimes *discuss* work with unions (for instance, his team at US West discussed the results of one study with telephone operators and union officers of

the Communication Workers of America; see [34]). But representatives within PICTIVE studies appear to be approached directly by researchers, not elected or selected by their constituencies. That's not surprising given the relative lack of labor organization within the US. In Scandinavia, unions are well organized and have an established organization for selecting representatives; in the US, so little of the workforce is unionized that work often happens in non-union shops.

Indeed, in sidestepping the unions, Muller found that his focus broadened to workers other than end users whose work lives would be affected, such as technical writers. Whereas the UTOPIA project partnered with a specific union, and consequently ran into conflicts of interest with other unions, PICTIVE led researchers to interact with workers of various stripes, often without affiliation to workers' organizations. For instance, one project involved home users of Microsoft software, users who included executives, home business people, and firefighters [35, pp.94-95].

PICTIVE's interpretation of the Scandinavian approach resembled cooperative prototyping: it "emphasizes a tradition of user participation in workplace decisions in improving the quality, productivity, and satisfaction related to computing systems" [39, p.212]. In PICTIVE, "users act as fully empowered participants in the design process" (212), which was based on a "partnership model" (213). The design that resulted represented a synthesis of different views (219), and in fact, "often, decisions are made by consensus rather than by voting or other approaches that might disenfranchise the minority" (220). So, although the ideal was *democratic*, PICTIVE often relied on *consensus* within meetings – an important distinction, since democracy assumed competing interests that were sometimes hard to reconcile. (Imagine deciding the results of elections or bond issues by consensus.) Appealing to consensus is more common in management teamwork approaches such as TQM (see [55]).

PICTIVE was a "microscopic"² design approach [51, p.51], focusing on individual screens or pages. Although a strong technique for supporting the development of systems, it assumed that the overall task is correct. But Leslie Tudor found that assumption to be suspect. To complement PICTIVE with a "macroscopic" approach that allowed workers to develop workflow, Tudor et al. developed CARD [51]. Together, PICTIVE and CARD provided a "bifocal" approach [42; see also 40] that allowed designers and workers to partner on both levels.

CARD originated as "an informal or semistructured card game that supports collaborative analysis and critique of a software system by a group of people [including users, developers, and designers] who have a variety of disciplines and skill sets" [51, p. 51]. Since then, it has been expanded to support other types of work [34, p.35]. Like PICTIVE, CARD could be used one-on-one or in groups. Like mock-ups and cooperative prototyping, CARD provided a common language for stakeholders.

As CARD developed, it became more sophisticated, allowing a layered approach that distinguished among "(1) observable,

¹ Although CARD is technically not a prototyping approach in the sense of mock-ups and cooperative prototyping ([51] compares it to the organizational toolkits of cooperative design), I include it here because it uses prototyping techniques and is tightly connected to PICTIVE.

² "Microscopic" is Muller's term. Elsewhere, I describe a taxonomy under which PICTIVE would be classified as "mesoscopic" [50].

formal components, (2) skill and craft, and (3) interpretive description” [35, p.90]. Its outcomes included “enhanced personal and organizational communication, improved teamwork, greater organizational cohesion, and stronger commitment by participants to the success of a project in which they (and, by extension, the constituencies that they represent) now have shared ownership” (91). Tudor et al. said that in one project, “co-creation and co-evaluation of new task flows, and shared responsibility for the resulting design, helped to reestablish trust in what had been a tense political situation” [51, p.52].

5.2 Comparison to the Scandinavian Challenge

Muller listed three goals for PICTIVE: to improve design by leveraging workers’ expertise; to improve the flow of the software engineering process by bringing in representative workers as co-owners of the design; and to empower workers to contribute to the process [38, 39]. These three motives of quality, commitment, and democracy were seen as convergent and mutually beneficial to employers and workers [39] – an interesting contrast to the Collective Resources Approach, in which researchers saw a fundamental, irreconcilable contradiction between capital’s and labor’s interests, and explicitly rejected the notion that the two could be balanced [7, p.84]. Unlike UTOPIAn mock-ups, but like cooperative prototyping, PICTIVE saw functional empowerment as leading to (or at least convergent with) democratic empowerment. Methodologically, PICTIVE represented a continuation of the shift in assumptions away from Marxist capital-labor conflict (the perpetual, irreconcilable struggle between higher profit margins and industrial democracy) and towards US-style capitalism (in which the interests of labor and capital are seen to converge, and conflicts between them are seen as the failure of either side to find common ground). The workforce was seen as divergent and involving multiple stakeholders, not divided between the two sides of capital and labor. And PICTIVE entailed bringing representatives of these different stakeholders together to explore each others’ perspectives [38, p.228].

To that end, CARD’s goals included providing an effective means of communication among stakeholders, leading to a shared responsibility for redesigned task flows and products based on them [51]. It represented a continuation of the exploration of tacit knowledge that originated with UTOPIAn mockups and became more central with cooperative prototyping: an understanding of *what* is done in work, but also *why* it is done [35]. The results are posters that could be viewed by all stakeholders, including management. Again, CARD represented a shift away from overtly political democratic empowerment and toward functional empowerment.

6. CONTEXTUAL DESIGN (US)

6.1 Description

Contextual design is a well-developed set of techniques for examining a workplace, gathering requirements, and developing solutions based on those requirements. Based in usability engineering efforts at Digital Equipment Corporation [52], contextual design represents the application of ethnographic methods and quality management tools to software engineering.

Prototyping is actually a rather small component of the contextual design process, just as in cooperative design (cf. [33]). First, researchers conduct short observations of work and interviews with representative users. (The users do not elect their own representatives; researchers typically approach them individually, and sometimes they are selected by managers.) Next, the design team reviews the observational notes and interviews and constructs several models of the work. These models are consolidated so that designers can get a macroscopic view of the work (see [50]). Based on these models, the design team creates a User Environment Description (UED), an outline of the new system to design. The UED leads to a series of low-fidelity prototypes.

Initially, these prototypes were used in a confirmatory rather than a cooperative fashion, in usability tests ([53, 54]; Beyer & Holtzblatt [5] later reject this approach). Later, prototypes were used in “prototyping interviews,” in which “the interviewer narrows the focus [of the investigation] considerably to find out what problems might exist with a storyboard or prototype” [4, p.412]. Finally, prototyping was adapted as an integral part of the contextual design process, even meriting its own chapter in Beyer and Holtzblatt’s book (see also [29, 46, 47]).

In [5], contextual design used prototypes to supply a common design language between design teams and customers. The common language was needed because customers cannot understand the design team’s specialized notations (p.369). Contextual design reinforced a division of labor between the design team (who constructed an overall picture of the work using specialized notations) and the users (who understood their own work at an individual level, tacitly and unreflectively).

Beyer and Holtzblatt have argued that “customers aren’t technologists – they don’t know the range of possibilities that technology could support. They may be either unrealistic or excessively cautious as a result. And they don’t know what it takes to make a design hang together. And why should they, after all? It’s their job to do their job, not to design systems” (p.371). Workers were shut out of the design process itself, although they were encouraged to give suggestions.

6.2 Comparison to the Scandinavian Challenge

Although Beyer and Holtzblatt presented prototyping as one technique for drawing the worker into the design process, and they even said that it involves making the worker a “co-designer,” they described prototyping as an avenue to functional empowerment. Indeed, the issue of democratic empowerment has not even made an appearance in contextual design literature as far as I can tell, even though it figured so prominently in the literature that Beyer and Holtzblatt cited when describing their prototyping approach. The issue of workplace democracy simply ceased to be an issue of concern.

But contextual designers continued to use some of the *language* of the Scandinavian approach. For instance, Beyer and Holtzblatt said that “we want to co-design the system with the users” [5, p.370], but their idea of co-design was quite different from the Scandinavian approach. They argued that prototyping was necessary because “no one articulates their work practice as an ordinary thing” (p.370); the Collective Resource Approach actually involved teaching workers to do just that – to critically

examine their work practices, explore their tacit knowledge, and determine how to integrate these with technology.

Designers were seen as best equipped to make design decisions and can be trusted to make them with the workers' good in mind. Beyer and Holtzblatt, then, wanted to provide *functional* empowerment. Workers were empowered to be more efficient, more productive, more flexible – in short, to meet management's goals! They were *not* empowered to redesign their work flow or control their own tools; designers had that prerogative because only they have the training to do so.

In contrast, the Scandinavian approach revolved around *democratic* empowerment, and a major component of the approach involved making workers into actual co-designers and teaching them the possibilities of integrating technology into their work. Workers were actually encouraged to *question* designers who had been hired by management, since in the Scandinavian approach these designers were unlikely to represent the interests of workers.

Contextual design's prototyping component, indeed, could be seen as the realization of the Scandinavians' worst fears. Repeatedly, the insights from prototyping were warranted in terms of profits and efficiency. For instance, when Beyer and Holtzblatt remarked that prototypes could help designers to spot emergent work practices, they emphasized that this understanding provided a *competitive advantage* (p.375). In another example, designers were told to ask commercially oriented questions at the end of the prototyping session: "Does he [sic] like it? Would he buy or recommend buying it? How much would he pay for it?" (p.400). Furthermore, prototyping was framed as primarily giving insight to the knowledgeable designers rather than the workers, who were not to be distracted with actual design work. For instance, Beyer and Holtzblatt said that the results of the prototyping session were useful data for the designer, but not actually a co-designed product: "As long as the user's intent is met, the UI designer is free to think up a better mechanism" (p.400).

The goals of contextual design, then, matched up with what the Scandinavian approach has labeled management's goals: efficiency, productivity, and flexibility (work intensification); increased competitiveness in the marketplace (the maximization of profits); and redesign with minimal interruption to work (separating decision-making from labor).

7. BUT HOW HAVE THE SCANDINAVIANS RESPONDED TO THE SCANDINAVIAN CHALLENGE?

Based on the declarations of Ehn, Kyng, and the other Scandinavians who issued the "Scandinavian challenge," it's easy to think that the Scandinavian approach is a strong democratic movement that has been corrupted with US-style capitalism as its prototyping techniques migrated to the US – that the Scandinavian challenge has been fumbled in transition. Indeed, contextual design in particular may seem to be the confirmation of Bansler and Kraft's fears that "US managers are likely to 'cherry pick' those aspects of the 'Scandinavian model' they find useful for their own ends and integrate them with familiar strategies" [3, p.103]. But it's worth noting that the Scandinavians themselves have backed away from the challenge they issued – even though, as Bansler and Kraft point out, their language of democratic

empowerment hasn't [3, p.98]. Later work, even from CRA firebrands such as Ehn and Kyng, has talked far less about the struggle between capital and labor and rarely invokes unions as partners (e.g., [24, 33]). Kyng [33] even includes managers as participants! Despite calls to revive the political agenda [7, 10], that agenda seems to have taken a back seat to the development of methods and techniques.

Why? Some link the change to the fact that despite the Marxist ideology of the CRA, Scandinavian unions never really bought into the antagonistic model that pitted capital against labor, preferring to see their relationship with management as cooperative [3, 32]. Others argue that the simple Marxist story of capital vs. labor, although perhaps valuable for critiquing the industrial capitalism of Marx's time, does not take into account the multidisciplinary nature of the modern workplace with its many competing stakeholders [7] and its blurred lines between management and workers [1]. Still others attribute the shift to a larger shift in Western Europe away from unions and manufacturing [20]. Whatever the case, later Scandinavian work tends to focus on functional rather than democratic empowerment. That is, the most important response to the Scandinavian challenge has been the Scandinavians': methodologically, they have crept ever closer to their compatriots in the US.

8. IMPLICATIONS FOR TECHNICAL COMMUNICATORS

What lessons can we as technical communicators draw from this comparison of Scandinavian and US approaches?

Technical communicators have often treated participatory design as a monolith – as long as users are somehow involved, it's participatory design ([11, 28, 31, 48]; see [44] for a counterexample). But human-computer interaction researchers have argued that participatory design also involves the relationship among designers and users and the manner in which the two interact [9]. In this paper, I've explored one participatory design technique, prototyping, and in the process I've differentiated between the assumptions of four different sets of methodological assumptions for participatory design. If technical communicators uncritically adopt a technique such as prototyping, we may find that it does not match our exigencies or ideology. And if we adopt a *range* of different techniques, we may find that they actually work *against* each other.

For instance, without the ideological apparatus and social conditions of Scandinavia in the early 1980s, UTOPIAn mockups do not make much sense in the US. They require unions to drive the process, workers who have been selected by unions as representatives, and a commitment to democratic improvement that works directly against employers' interests. They presume a conflict between capital and labor and are meant to be weapons in that struggle. And they are emphatically not meant to make the workplace more efficient. These conditions in turn require long-term employment, which is to say low career mobility, so that employers and workers share a long-term commitment to each other. Implementing UTOPIAn mockups in a non-unionized workplace, one in which workers do not presume a struggle with employers, would not only be difficult but counterproductive – to the interests of all involved. Imagine applying this time-consuming, work-disrupting technique in conjunction with contextual design, which has been adapted to the time and

resource constraints of engineering in US companies [30]! The UTOPIAn mock-up technique would have to be severely revised – some would say co-opted – to work with contextual design.

My hope is that this paper can help technical communicators to find a “best fit” prototyping technique for their current situation. But it can also help them to develop new techniques that are a better fit. They might examine their own ideological goals and work situation in this light.

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